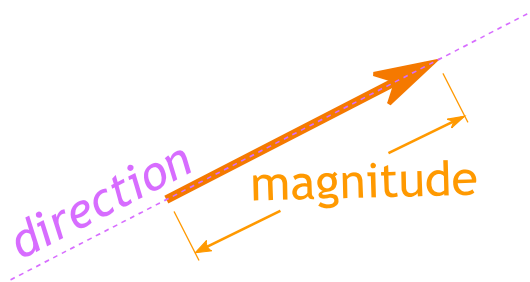


Dot Product

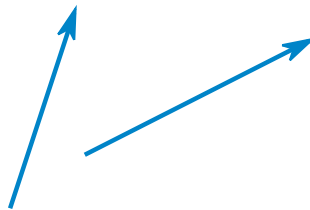
We may use [Cookies](#)

OK

A [vector](#) has **magnitude** (how long it is) and **direction**:



Here are two vectors:



They can be **multiplied** using the "**Dot Product**" (also see [Cross Product](#)).

Calculating

The Dot Product is written using a central dot:

$$\mathbf{a} \cdot \mathbf{b}$$

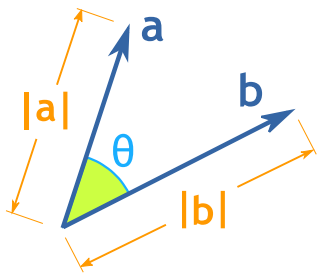
This means the Dot Product of **a** and **b**

We can calculate the Dot Product of two vectors this way:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \times |\mathbf{b}| \times \cos(\theta)$$

Where:

$|\mathbf{a}|$ is the magnitude (length) of vector **a**

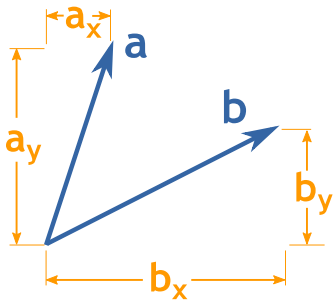


$|\mathbf{b}|$ is the magnitude (length) of vector $\mathbf{b}$
 θ is the angle between $\mathbf{a}$ and $\mathbf{b}$

So we multiply the length of $\mathbf{a}$ times the length of $\mathbf{b}$, then multiply by the cosine of the angle between $\mathbf{a}$ and $\mathbf{b}$

OR we can calculate it this way:

$$\mathbf{a} \cdot \mathbf{b} = a_x \times b_x + a_y \times b_y$$

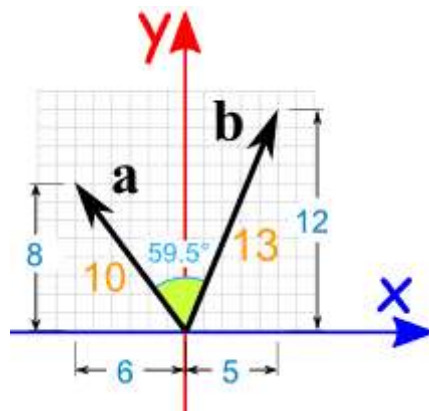


So we multiply the x's, multiply the y's, then add.

Both methods work!

And the result is a **number** (called a "scalar" to show it is not a vector).

Example: Calculate the dot product of vectors $\mathbf{a}$ and $\mathbf{b}$:



$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \times |\mathbf{b}| \times \cos(\theta)$$

$$\mathbf{a} \cdot \mathbf{b} = 10 \times 13 \times \cos(59.5^\circ)$$

$$\mathbf{a} \cdot \mathbf{b} = 10 \times 13 \times 0.5075...$$

$$\mathbf{a} \cdot \mathbf{b} = 65.98... = 66 \text{ (rounded)}$$

OR we can calculate it this way:

$$\mathbf{a} \cdot \mathbf{b} = a_x \times b_x + a_y \times b_y$$

$$\mathbf{a} \cdot \mathbf{b} = -6 \times 5 + 8 \times 12$$

$$\mathbf{a} \cdot \mathbf{b} = -30 + 96$$

$$\mathbf{a} \cdot \mathbf{b} = 66$$

Both methods came up with the same result (after rounding)

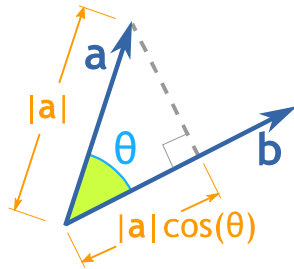
Also note that we used **minus 6** for a_x (it is heading in the negative x-direction)

Note: you can use the [Vector Calculator](#) to help you.

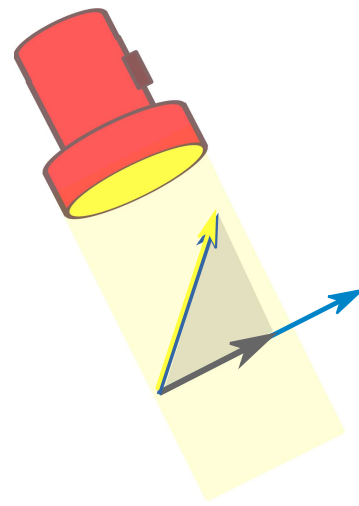
Why $\cos(\theta)$?

OK, to multiply two vectors it makes sense to multiply their lengths together **but only when they point in the same direction.**

So we make one "point in the same direction" as the other by multiplying by $\cos(\theta)$:

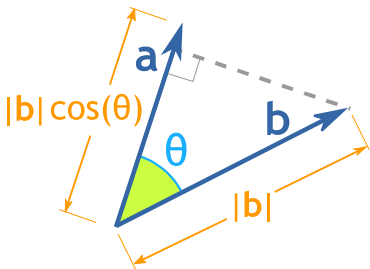


We take the component of **a** that lies alongside **b**



Like shining a light to see where the shadow lies

THEN we multiply !



It works exactly the same if we "projected" **b** alongside **a** then multiplied.

Because it doesn't matter which order we do the multiplication:

$$|\mathbf{a}| \times |\mathbf{b}| \times \cos(\theta) = |\mathbf{a}| \times \cos(\theta) \times |\mathbf{b}|$$

In effect, the dot product multiplies the **aligned** lengths.

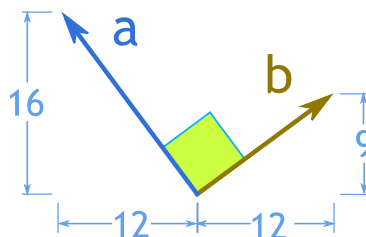
Remembering Cos

To remember to multiply by $\cos(\theta)$ think "dot cos".

Right Angles

When two vectors are at right angles to each other the dot product is **zero**.

Example: calculate the dot product for:



$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \times |\mathbf{b}| \times \cos(\theta)$$

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \times |\mathbf{b}| \times \cos(90^\circ)$$

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \times |\mathbf{b}| \times 0$$

$$\mathbf{a} \cdot \mathbf{b} = 0$$

or we can calculate it this way:

$$\mathbf{a} \cdot \mathbf{b} = a_x \times b_x + a_y \times b_y$$

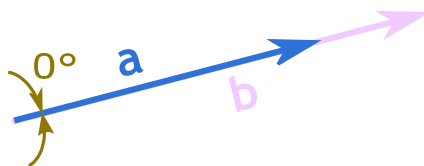
$$\mathbf{a} \cdot \mathbf{b} = -12 \times 12 + 16 \times 9$$

$$\mathbf{a} \cdot \mathbf{b} = -144 + 144$$

$$\mathbf{a} \cdot \mathbf{b} = 0$$

This can be a handy way to find out if two vectors are at right angles.

Same Direction



The dot product of two vectors that point in the same direction is the simple product of their lengths, because the angle is 0 degrees which has a cosine of 1

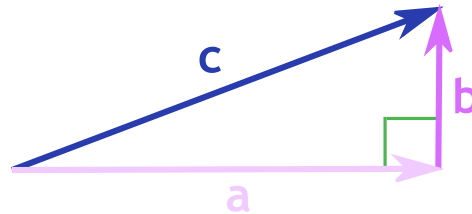
$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \times |\mathbf{b}| \times \cos(0^\circ)$$

$$\mathbf{a} \cdot \mathbf{b} = a \times b \times 1$$

$$\mathbf{a} \cdot \mathbf{b} = ab$$

Right-Angled Triangle

Let's use the dot product on a right-angled triangle!



Vector **c** is the sum of **a** and **b**: $\mathbf{c} = \mathbf{a} + \mathbf{b}$

Do a dot product on both sides: $\mathbf{c} \cdot \mathbf{c} = (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b})$

Expand: $\mathbf{c} \cdot \mathbf{c} = \mathbf{a} \cdot (\mathbf{a} + \mathbf{b}) + \mathbf{b} \cdot (\mathbf{a} + \mathbf{b})$

Expand Again: $\mathbf{c} \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{a} + \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{a} + \mathbf{b} \cdot \mathbf{b}$

$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a} = 0$ (right angles): $\mathbf{c} \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{a} + \mathbf{b} \cdot \mathbf{b}$

$\mathbf{c} \cdot \mathbf{c} = c^2$ etc: $c^2 = a^2 + b^2$

We just proved the [Pythagorean Theorem](#)!

Note: when we allow angles other than 90 degrees we can create the [Law of Cosines](#). Have a go yourself, but be careful how you define the angle!

Physics

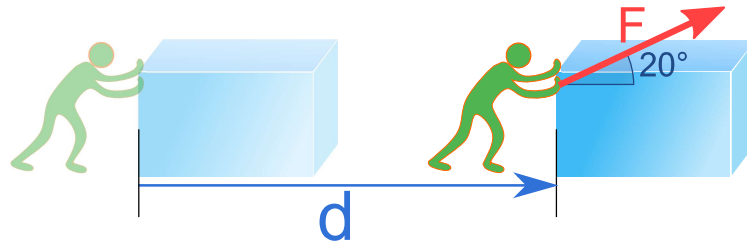
The Dot Product is used a lot in Physics

Example: Work

In Physics **Work** is force times distance, but only the **aligned** part.

So work is the dot product of force and distance.

Alex pushes a box 3 m straight forward using 200 N of force. But his push is a little upwards by 20° .



$$\begin{aligned}
 \text{Work} &= \text{Force} \cdot \text{Distance} \\
 &= \text{Force} \times \text{Distance} \times \cos \theta \\
 &= 200 \text{ N} \times 3 \text{ m} \times \cos 20^\circ \\
 &= 200 \text{ N} \times 3 \text{ m} \times 0.9397... \\
 &= 564 \text{ J (to nearest Joule, where J = N m)}
 \end{aligned}$$

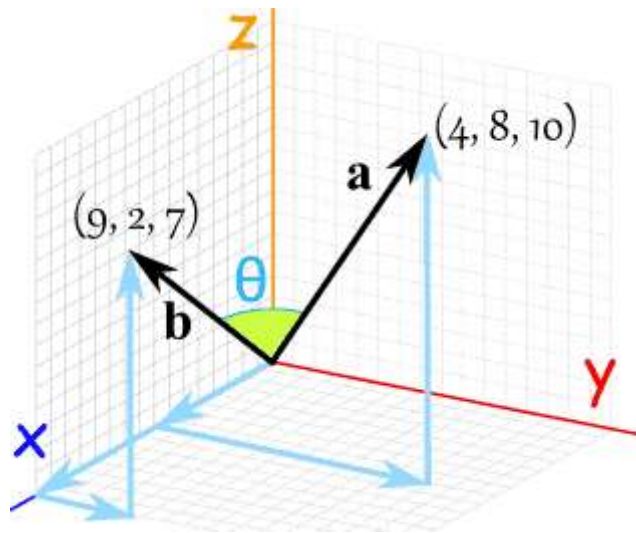
(Without $\cos \theta$, the **wrong** value would be 600 J)

Three or More Dimensions

This all works fine in 3 (or more) dimensions, too.

And can actually be very useful!

Example: Sam has measured the end-points of two poles, and wants to know **the angle between them**:



We have 3 dimensions, so don't forget the z-components:

$$\mathbf{a} \cdot \mathbf{b} = a_x \times b_x + a_y \times b_y + a_z \times b_z$$

$$\mathbf{a} \cdot \mathbf{b} = 9 \times 4 + 2 \times 8 + 7 \times 10$$

$$\mathbf{a} \cdot \mathbf{b} = 36 + 16 + 70$$

$$\mathbf{a} \cdot \mathbf{b} = 122$$

Now for the other formula:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \times |\mathbf{b}| \times \cos(\theta)$$

But what is $|\mathbf{a}|$? It is the magnitude, or length, of the vector $\mathbf{a}$. We can use

Pythagoras:

- $|\mathbf{a}| = \sqrt{4^2 + 8^2 + 10^2}$
- $|\mathbf{a}| = \sqrt{16 + 64 + 100}$
- $|\mathbf{a}| = \sqrt{180}$

Likewise for $|\mathbf{b}|$:

- $|\mathbf{b}| = \sqrt{9^2 + 2^2 + 7^2}$
- $|\mathbf{b}| = \sqrt{81 + 4 + 49}$

- $|\mathbf{b}| = \sqrt{134}$

And we know from the calculation above that $\mathbf{a} \cdot \mathbf{b} = 122$, so:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \times |\mathbf{b}| \times \cos(\theta)$$

$$122 = \sqrt{180} \times \sqrt{134} \times \cos(\theta)$$

$$\cos(\theta) = 122 / (\sqrt{180} \times \sqrt{134})$$

$$\cos(\theta) = 0.7855...$$

$$\theta = \cos^{-1}(0.7855...) = 38.2...^\circ$$

Done!

I tried a calculation like that once, but worked all in angles and distances ... it was very hard, involved lots of trigonometry, and my brain hurt. The method above is much easier.

Cross Product

The Dot Product gives a **scalar** (ordinary number) answer, and is sometimes called the **scalar product**.

But there is also the [Cross Product](#) which gives a **vector** as an answer, and is sometimes called the **vector product**.

Mathopolis: [Q1](#) [Q2](#) [Q3](#) [Q4](#) [Q5](#) [Q6](#) [Q7](#) [Q8](#) [Q9](#) [Q10](#)